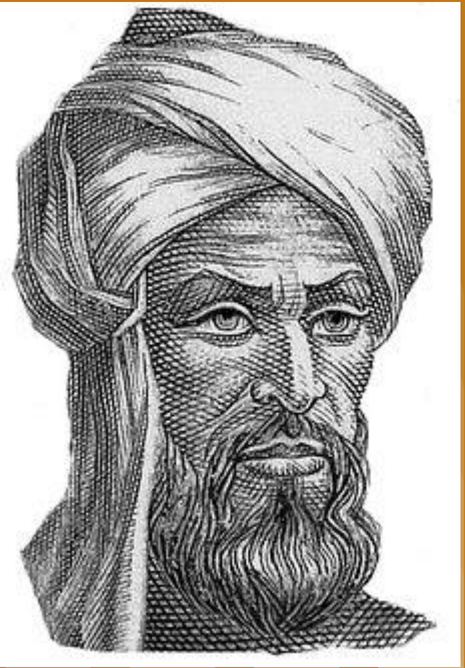
Muḥammad ibn Mūsā al-Khwārizmī

Daniel al-Correia



The Legend

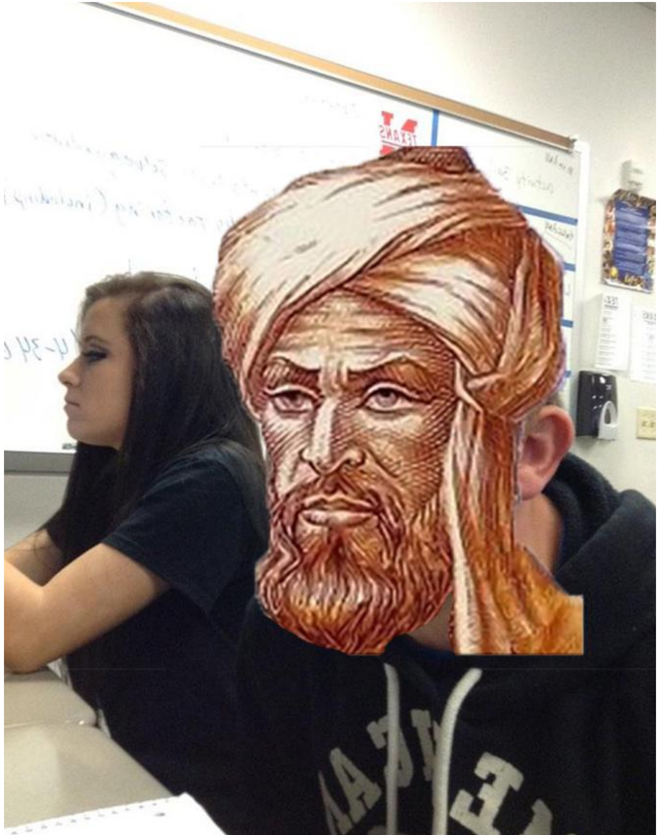
Muḥammad ibn Mūsā al-Khwārizmī, born around 780, was a Muslim mathematician whose works mainly introduced Hindu-Arabic numerals and concepts of algebra solving formulas into European mathematics [1].

Al-Khwarizmi was a Persian man, most likely born somewhere in Central Asia near today's Uzbekistan [2].

A man of a thousand talents

Al-Khwarizmi was one of the learned men who worked in the House of Wisdom in Baghdad [2].

He was part of a community of scholars from across the world who translated and studied ancient manuscripts on science, math, medicine, history, philosophy, and more [3].

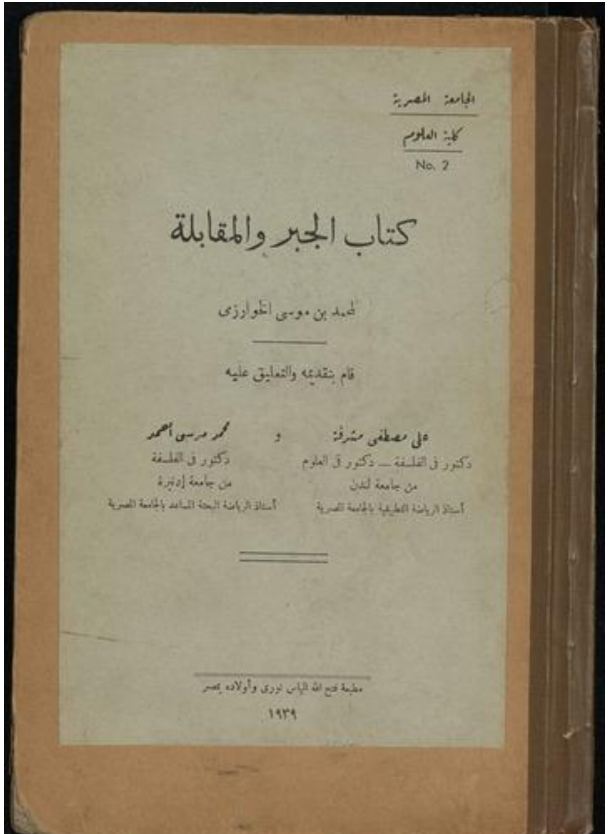


al-Kitāb al-mukhtasar fīhisāb al-jabr wal-muqābala

He improved techniques we use to solve algebraic problems. Even though he did not invent algebra, he developed the way we see mathematical problems.

This book offered detailed instructions for solving linear and quadratic equations, earning him the title “father of algebra.”

(The Compendious Book on Calculation by Completion and Balancing) [3]



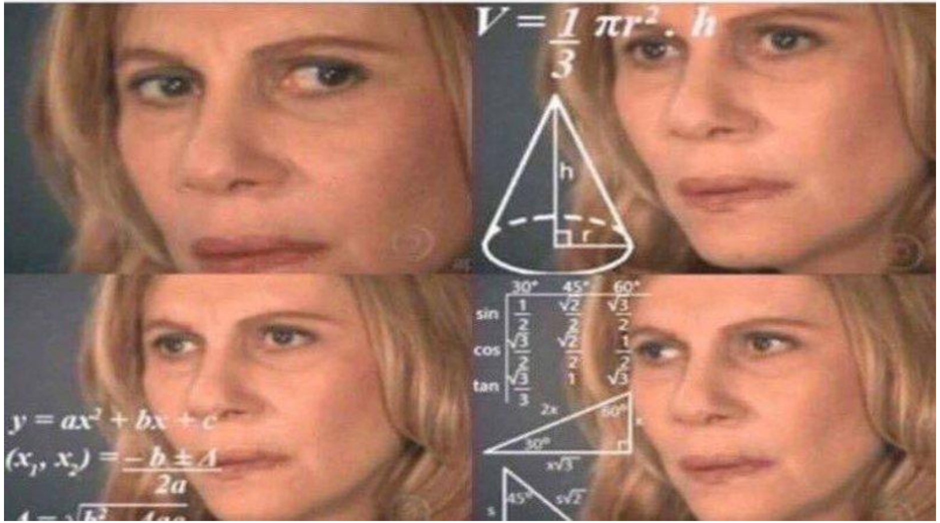
Solving first world problems

His book explained how to use equations to solve real life problems, like how to split an inheritance, divide a plot of land, and find measurements for canals and buildings [3].

The rules to solve a problem

While al-Khwarizmi was not the first person to understand these equations, he was the first to provide algorithms for solving them [2].

Algorithms are sets of rules to solve a problem.



Visionary

al-Khwarizmi explained how to solve equations in words.

ax2 + bx + c = 0

Although he had this approach, his second book encouraged mathematicians to use the Hindu numbering system [3].

Hindu-Arabic numerals

Al-Khwarizmi popularized the Hindu-Arabic system, developed in ancient India, in the Islamic world and his book is responsible for their adoption in Europe centuries later [2].

The system had nine symbols to represent numbers and a circle around empty space to represent “nothingness.”

To keep from having to use more symbols for bigger numbers, the Hindu system was a place system. The number could be determined by its place in a row of numbers: There was a row for 1s, 10s, 100s, 1000s, … [4].

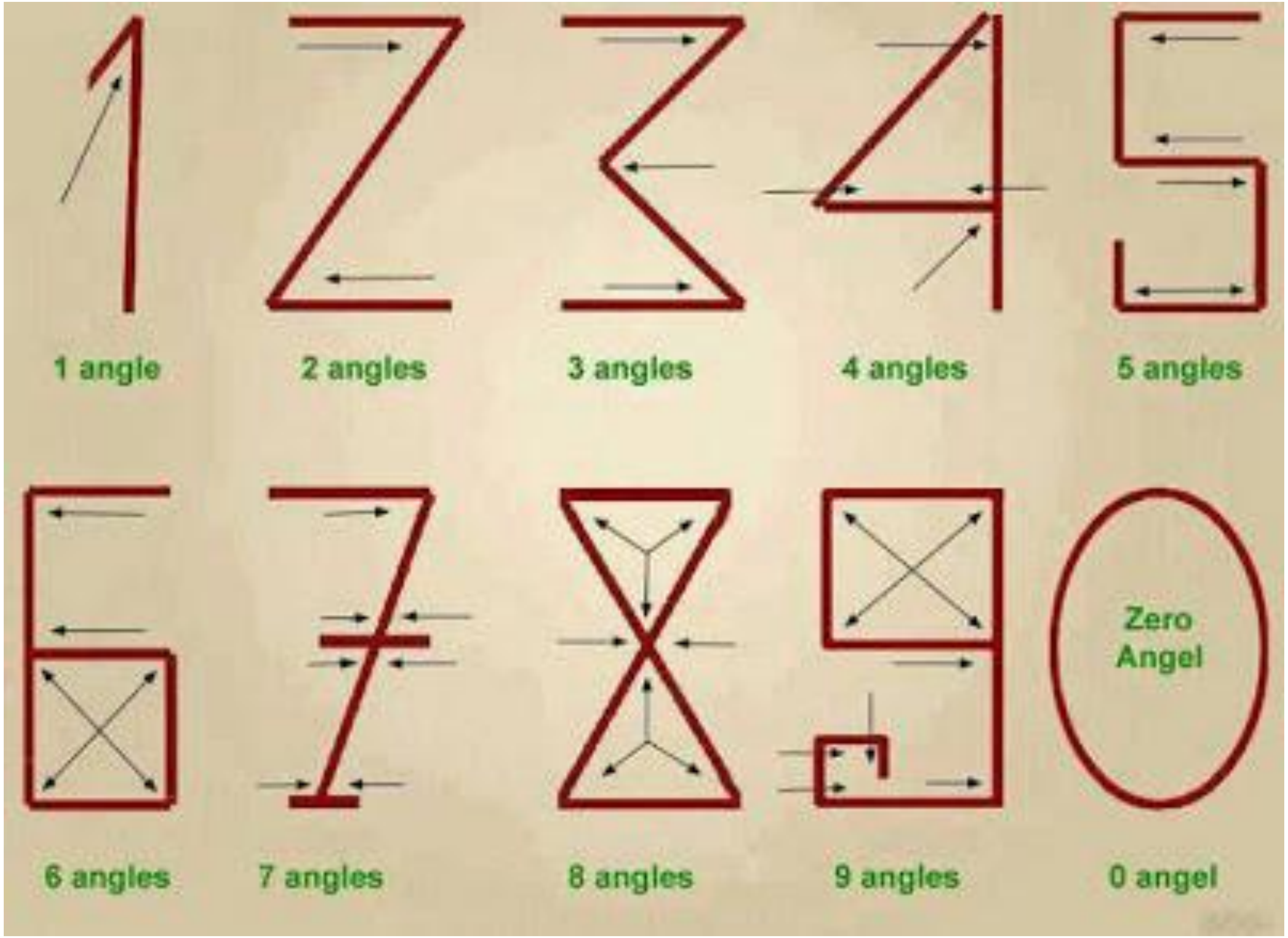
Eureka!

The Hindu circle remained a placeholder.

3 – 4 =___

...2, 1, -1, -2...the fourth number was -2

Between +1 and -1


1 angle
2 angles
3 angles
4 angles
5 angles
Zero
Angel
6 angles
7 angles
8 angles
9 angles
0 angel

Sources

[1] www.britannica.com

[2] www.famousscientists.org

[3] www.khanacademy.org

[4] www.todayifoundout.com

Thanks for your attention